HW 6

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ECEN 164

1. $I_L = 80A$, Images of graphs on HW pdf

a.

i. inductor (output filter)

ii. $Z = j\omega L$ (higher L, higher Z)

Higher

$\Delta I_L = \frac{(V_{in} - V_{out})D}{L f_s}$; small L → big ripple

iii.

red (L↓)	green (L↑)
advantages: less losses, less area, less cost	advantages: less ripple (V_L and I_L)
disad: more ripple on V_L and I_L	disad: costs more, more area, more losses

b.

i. capacitor (output filter)

ii. $\Delta V_{out} = \frac{\Delta I_L}{8 f_s C}$; Large C → Less ΔV_{out}

$Z = \frac{1}{j\omega C}$; Lower C = higher imp

iii.

Black (#3)	Blue (#4)
ad: higher C → lower Z → Less ΔV_{out} ripple	ad: lower C → cheap → small
dis: more expensive, greater area	dis: greater ripple, higher Z, deeper impedance dip

2. $f_s = 200K$; Need 9W out 12V, from $V_{in} = 3V$, $T_s = 5\mu s$

$V_o = \frac{V_s}{1-D} \rightarrow 12 = \frac{3}{1-D} \rightarrow D = .75$ $\quad I_o(12V) = 9 \quad I_o = .75A$, $I_{in} = 3A$

$\Delta I_{LON} = \frac{V_s * DT_s}{L} \rightarrow .3 = \frac{3(.75)(5\mu s)}{L} \rightarrow L = 37.5mH$

$\frac{\Delta I_{LON}}{I_o} \leq .1 \rightarrow I_{LON} \leq .3$

$\frac{12}{.75} = R = 16\Omega$

$\frac{\Delta V_o}{V_o} = \frac{D}{RCf}$

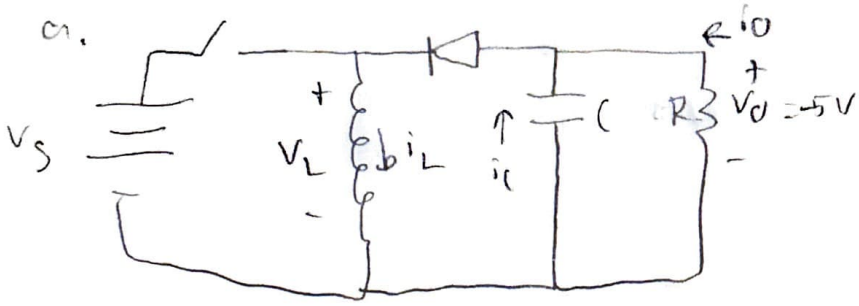
$.005 = \frac{.75}{16C(200K)}$

$C = 46.875 mF$

$\boxed{\begin{array}{l} C = 46.875\ mF \\ R = 16\Omega \\ L = 37.5mH \end{array}}$

3.

a.



b.

$R = 4\Omega$

$D = .25$

$-15\left(\frac{.25}{1-.25}\right) = \boxed{V_O = -5V}$

c.

BB avg $I_L = I_{in} + I_O$

$\frac{5}{4} = 1.25 = I_O$

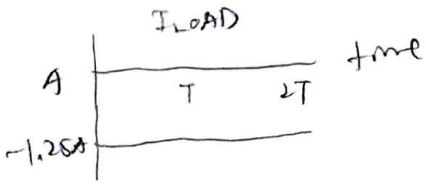
$5(1.25) = 15(I_{in})$

$I_{in} = .4167A$

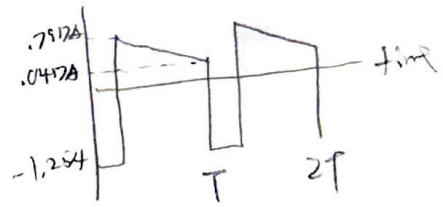
$\boxed{I_L = .4167 + 1.25 = 1.667A}$

d.

$I_{Load} = -1.25A$



I_C



$\frac{.25}{2} + 1.667 - 1.25 = .7917A$

$1.667 - \frac{.25}{2} - 1.25 = .0417A$

e.

L

Highest voltage = 15V

Highest current $= 1.667 + \frac{.25}{2} = 2.042A$

$\boxed{\text{15V and 2.042A Inductor ratings}}$

C

Highest voltage $= |V_O| = 5V$

Highest current $= |I_O| = 1.25A$

$\boxed{\text{5V and 1.25A Capacitor Ratings}}$